

Sumanth Kumar Meesala

Senior AI Engineer · LLM & Agent Applications · AI-Native Full-Stack

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SUMMARY

Senior AI Engineer with 11+ years building production Node.js, React, and Angular systems — now an AI-native product engineer shipping LLM-powered features, autonomous agents, and multi-step agentic workflows on AWS. At Affle: core contributor to Blueprix (an MCP-enabled agent-orchestration platform), owner of Qrank (360° review platform, 600+ employees globally), and mentor to the AI team; part of the team that built Moose (acquired by Kaluza, 2025). Daily driver of Claude Code, Cursor, GitHub Copilot, and Augment Code — I treat AI agents as engineering teammates with explicit responsibilities, context, and guardrails. Owns 0→1 prototypes through to hardened, tested, observable production systems.

SKILLS

AI-Native Engineering	Generative AI & LLM application development, Retrieval-Augmented Generation (RAG), Agentic AI, multi-agent systems & orchestration, tool/function calling, structured output, LLM evaluation (evals, LLM-as-judge), prompt & context engineering, guardrails
AI Tooling	Claude Code, Cursor, GitHub Copilot, Augment Code, MCP (Model Context Protocol) servers, autonomous coding agents, AGENTS.md / CLAUDE.md context engineering, N8N
LLM Stack	OpenAI GPT-4o, Anthropic Claude (Sonnet / Opus), Llama, Hugging Face, Amazon Bedrock (Converse, InvokeModel, Titan Embeddings, Guardrails), LangChain, LangGraph, LlamaIndex, CrewAI, vector databases: Pinecone, ChromaDB, FAISS, pgvector
Languages	TypeScript, JavaScript (ES6+), Node.js, Python, C#, HTML5, CSS
Full-Stack	React 18, Next.js, Redux / Zustand, Tailwind, Material UI, Angular, Express.js, REST / API design & integration, WebSockets, SSE, event-driven Node.js microservices, distributed systems
Cloud & DevOps	AWS (Lambda, EC2, API Gateway, S3, CloudFront, EventBridge, SQS, ECS, RDS, IAM), AWS CDK, CloudFormation, Serverless Framework, Infrastructure as Code, Docker, CI/CD: GitHub Actions, GitLab CI, Bitbucket Pipelines, Git
Data	PostgreSQL, DynamoDB, MongoDB, SQL Server, MySQL, Supabase, schema design & migrations
Testing	Jest, Vitest, Playwright, unit / integration / end-to-end testing, TDD
Security & Responsible AI	OAuth 2.0, JWT, RBAC, least-privilege IAM, Secrets Manager / SSM, Bedrock Guardrails, PII redaction, prompt-injection defenses
Observability	CloudWatch, structured logging, metrics, tracing, Langfuse
Leadership	Mentoring, code review, stakeholder management, Agile / Scrum, system & software architecture, technical leadership, JIRA

EXPERIENCE

Affle — Melbourne

Senior AI Engineer

Jul 2026 – Present

- Core contributor to Blueprix, a state-machine-based, MCP-enabled agent-orchestration platform — built its MCP servers/tool integration, multi-agent orchestration, checkpoint/resume state handling, and RAG context management plus agent UIs; standardised model access on Amazon Bedrock (Converse / InvokeModel streaming, Titan Embeddings, Guardrails for content and PII filtering).
- Own and operate Qrank (qrank.it.com), an LLM-powered multi-tenant 360° review platform used by 600+ employees across Affle's global offices — turning peer/manager/self feedback into structured review reports.
- Own Blueprix's CI/CD — GitHub Actions and Bitbucket Pipelines build, test, and deploy agent services to AWS, with pre-commit linting, secret scanning, least-privilege IAM, and secrets via SSM gating every release.
- Built the evaluation and observability loop — LLM-as-judge evals, prompt versioning, and tracing (Langfuse + CloudWatch) with cost and latency controls — so agent releases are gated on eval regressions.
- Mentor engineers across the AI team on agent design, prompt/context engineering, and AI-tooling adoption (Claude Code / Cursor onboarding, shared CLAUDE.md standards).

- Architected enterprise RAG pipelines (LangChain + LlamaIndex on Node.js / Python services) over technical documentation — ~40% retrieval-efficiency gain; multi-day analyst lookups replaced with sub-minute answers.
- Designed and ran autonomous AI agents (LangGraph, CrewAI) automating analytics, reporting, and DynamoDB orchestration; delivered bespoke multi-step agentic workflows for SMB / enterprise clients replacing manual SOP-driven back-office tasks.
- Integrated Claude and GPT-4o into event-driven Node.js microservices on AWS Lambda, EventBridge, and SQS with retry / guardrail layers, structured outputs, and prompt caching — reduced hallucination incidents and stabilised production cost.
- Owned CI/CD across GitHub, GitLab, and Bitbucket — pipeline templates, branch strategies, and automated multi-environment deploys for a multi-client delivery team.
- Introduced automated testing across services — Jest unit/integration suites and Playwright end-to-end tests wired into CI, blocking merges on failures.
- Co-built Moose (beige.tech/solutions/moose) — enterprise product platform on Angular + AWS serverless; owned full-stack delivery and provisioned infrastructure as code with AWS CDK (TypeScript) and Serverless Framework (Lambda, API Gateway, DynamoDB, EventBridge stacks). Acquired by Kaluza in 2025.
- Supported and enhanced Moonee Valley Council systems — a regulated public-sector environment — on C#, .NET Core, Entity Framework, and SQL Server; managed Azure/IIS-hosted application deployments across the broader .NET stack.
- Designed and built Qrank (qrank.it.com), a 360° review platform on Angular + Node.js + AWS with LLM-generated review reports — later deployed at scale across Affle (600+ employees).
- Mentored developers through code review, pairing, and design reviews; productionised Claude Code, Cursor, and Augment Code as the team default (shared CLAUDE.md / AGENTS.md, custom slash commands) — ~30% faster feature cycle.
- Led the React 18 + TypeScript re-architecture replacing legacy Angular modules; built real-time dashboards over WebSocket streams.

- Translated business requirements into MERN-stack applications (MongoDB, Express, React, Node.js) for B2B clients; owned features end-to-end from spec to production.
- Designed responsive React + Redux interfaces with reusable component patterns; standardised form validation, error handling, and loading states.
- Built REST APIs with JWT auth, role-based access control (RBAC), and audit logging; wrote MongoDB aggregation pipelines and indexes for read-heavy workloads.
- Integrated third-party services (payment, email, file storage) behind clean adapter layers; participated in technical-design reviews and code reviews.

- Built React frontends bridged to Node.js + MongoDB backend services; converted design specs into production-ready, accessible components.
- Implemented OAuth 2.0 authentication and authorization with secure session handling, refresh-token rotation, and protected routes.
- Wrote modular Express middleware for request validation, rate limiting, and structured error responses.

- Engineered single-page applications using React and Redux integrated with RESTful Node.js / Express endpoints backed by MongoDB.
- Built domain list/detail flows, search, and filtering with focus on perceived performance — skeleton loaders, optimistic updates, paginated lists.
- Authored API services with consistent input validation, error contracts, and logging.

- Delivered .NET enterprise web applications (ASP.NET MVC, AngularJS, WCF, Entity Framework, C#, HTML5, JavaScript) across the full SDLC in Agile/Scrum teams for multiple client engagements.
- Designed data-access layers and optimised SQL Server queries with LINQ, indexed views, and stored procedures to meet enterprise SLA requirements.
- Implemented authentication / authorization (Forms, Windows, claims-based) and integrated with third-party SOAP and REST services.
- Reviewed pull requests, mentored and ramped junior developers on the codebase, and maintained engineering standards across release cycles.

EDUCATION